CertiVar: Prevalence-Free Certified Assay Likelihood Ratios for Missense Variant-Effect Predictor Calls

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Abstract

Variant-effect predictors are a major source of evidence for missense variants of uncertain significance, and the Bayesian form of the ACMG/AMP guidelines expresses evidence strength as a likelihood ratio. Gene-specific calibrations estimate this ratio from few labelled variants, as point estimates or resampling-based bounds that carry no finite-sample guarantee once a threshold has been chosen from the same data. We present CertiVar, which turns a predictor score into loss-of-function (LoF) calls, functional calls or abstentions using class-conditional conformal thresholds learned from saturation genome editing or deep mutational scanning data, with one variant per residue because substitutions at a residue are dependent. It then certifies a bound on the likelihood ratio of each call type with respect to the assay outcome, using a Bonferroni correction or fixed-sequence testing so that the bound stays valid after a threshold is chosen from a grid. Every guarantee conditions on the assay class (per-class coverage at a fixed miscoverage level), so none requires the unknown share of LoF variants among the variants being called, provided that their within-class score distribution matches that of the assayed variants. On 14,477 assayed missense variants in BRCA1, BRCA2, BAP1, RAD51C and MSH2, scored by AlphaMissense, REVEL and ESM-2, worst-class coverage was 0.896–0.908 at a 0.90 target in all 15 gene–predictor pairs, and stayed at 0.89 or above when the LoF share of the calibration data was moved from 5% to 50%; marginal conformal prediction calibrated at a 5% share covered LoF variants at only 0.11 in its worst pair. In resampling experiments with known true likelihood ratios, every certified bound in a draw held in 99.8% of draws, against 56.5% for percentile-bootstrap bounds, and fixed-sequence testing more than doubled the certification of moderate LoF evidence. The certificates concern the assay, not pathogenicity; certified evidence was mostly supporting-level, needed same-gene assay data, and held in 88.8% of cases in held-out protein regions, below the nominal 95%. Code: https://github.com/aaryavvatts-lab/certivar.

Keywords: variant effect prediction; conformal prediction; likelihood ratio; ACMG/AMP evidence calibration; saturation genome editing; deep mutational scanning; variants of uncertain significance; multiple testing.

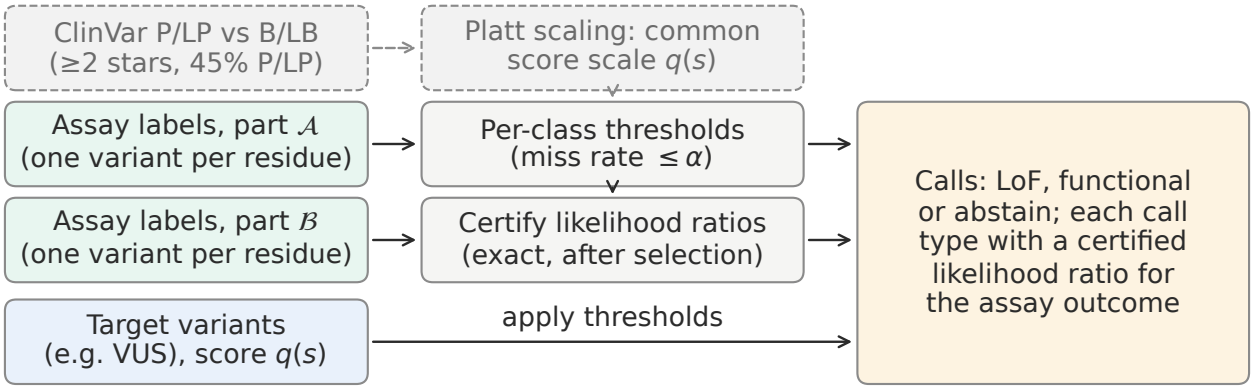


Figure 1: CertiVar. Per-class thresholds are set on part $\mathcal{A}$ of a gene's assayed residues, and the likelihood ratio of each call type with respect to the assay outcome is certified on a disjoint part $\mathcal{B}$; the thresholds are then applied to the target variants. The grey path (Platt scaling on ClinVar labels) only puts predictors on a common scale; it is a monotone map that does not change any call.

1 Introduction

In the five cancer-predisposition genes studied here, 81% of the ClinVar missense records classified as (likely) pathogenic, (likely) benign or uncertain are variants of uncertain significance (VUS) [1], and for many of them computational predictors are among the few sources of evidence [2]. The guidelines of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) grade evidence qualitatively [3]; their Bayesian adaptation equates each strength with a likelihood ratio, the odds of pathogenicity (OddsPath) [4, 5], so the strength of a predictor's evidence is a statistical estimate. Calibration workflows report it as a point estimate or a resampling-based bound [2, 6, 7]. When the calibration is specific to a gene, few labelled variants are available and the threshold that defines the evidence is chosen from the same data; none of these bounds carries a finite-sample guarantee that holds after this choice.

Two features of the data make the estimate fragile. First, the share of damaging variants (the prevalence) differs between calibration sets and the variants being classified, and is unknown for the latter: in our five genes, 45% of confidently labelled ClinVar missense variants are pathogenic, whereas 6.0% to 25.9% of the other assayed missense variants lose function in saturation genome editing (SGE) or deep mutational scanning (DMS) experiments (Table 1). Thresholds that depend on the class balance of the calibration set inherit this mismatch: calibrated on assay data at a 5% LoF share, marginal conformal thresholds covered the LoF variants of one gene–predictor pair at only 0.11 (Section 4.2). Second, gene-level labelled sets are small and substitutions at one residue are dependent: in a resampling design with known truth, percentile-bootstrap bounds were respected in only 76.5% of threshold selections, and evidence levels claimed from point estimates held in only 79.7% of claims (Section 4.4).

We introduce CertiVar (Figure 1), which turns a predictor score into loss-of-function (LoF), functional or no calls. Per-class (Mondrian) conformal thresholds set from SGE or DMS labels bound how often each class is mis-called, and a certification step bounds the likelihood ratio of each call type with respect to the assay outcome, valid after a threshold has been selected; neither needs an estimate of the LoF share. The certified quantity is how strongly a call predicts the assay result; it is evidence about pathogenicity only to the extent that the assay is clinically validated.

Our contributions are as follows. (1) We formulate predictor-based calling as abstaining classification with per-class error control on residue-clustered data, and prove that the certified likelihood-ratio bounds hold after selection from a grid of thresholds, that per-class coverage after the selection is at least one minus the largest grid level, and that a fixed-sequence variant certifies a level without a multiplicity correction over the grid (Section 3). (2) In a known-truth resampling design, exact binomial bounds on one variant per residue held where bootstrap bounds and point estimates often did not, and the fixed-sequence variant certified moderate evidence more than twice as often as the Bonferroni correction (Section 4.4). (3) On five genes and three predictors, per-class coverage stayed at the target when the calibration LoF share ranged from 5% to 50%, where marginal and prior-reweighted conformal prediction failed, and held approximately

on assayed VUS (Sections 4.2 and 4.3). (4) We quantify the cost and the limits: mostly supporting-level evidence, abstention on about a third of variants or more, the need for same-gene assay data, and weaker certificates in held-out protein regions (Section 4.5).

2 Related work

Calibrating predictors to clinical evidence. ClinGen maps predictor scores to ACMG/AMP evidence strengths through local likelihood ratios estimated against ClinVar labels, with a one-sided bootstrap bound and a prior estimated by a positive–unlabelled method [2]; the approach has been extended to more predictors [6], to assay scores [5, 8, 9], and to gene- and domain-specific calibration [7, 10, 11]. Where these report uncertainty, it comes from resampling: the bootstrap [2, 6] or sign-test bounds on the median of resampled likelihood ratios [7]. Such bounds are approximate at small sample sizes and are not adjusted for the choice of a threshold from the same data.

Conformal prediction, selective classification and risk control. Split conformal prediction gives finite-sample coverage [12]; its class-conditional (Mondrian) form controls coverage within each class [12, 13]; order-statistic thresholds give high-probability, training-conditional control [14, 15]; and keeping one member per group handles clustered data [16]. Abstaining classifiers trade the share of inputs they classify against error [17, 18]. Selecting a threshold from a grid with binomial bounds and a multiplicity correction, or by fixed-sequence testing, is the template of risk-controlling prediction sets and Learn-then-Test [19, 20]. Harrizi et al. reported split-conformal prediction sets for missense predictors on ClinVar benchmarks at 90% marginal coverage, treating a set with both labels as a refusal to call [21]; CertiVar instead controls coverage within each class, calibrates on the gene's assay labels and certifies the likelihood ratio of the calls. Our certificate divides two exact one-sided Clopper–Pearson endpoints; log-scale and score-type intervals for a ratio of binomial proportions [22, 23] are usually narrower but only asymptotically valid, which matters at the counts of a single gene. The building blocks are known; what is new is their combination into a certificate on the OddsPath scale for residue-clustered assay labels, with a coverage guarantee after threshold selection, a fixed-sequence variant, and a known-truth benchmark of how often point estimates and bootstrap bounds overstate evidence. Label-shift methods instead adjust posteriors or conformal weights to an estimated target prior [24–26]; class-conditional quantities need no prior, provided the target satisfies class-conditional invariance (Assumption 2), which we test.

3 Method

Setting. Each missense variant has a predictor score s, oriented so that higher means more damaging, and an assay label $Y \in \{0, 1\}$, with $Y = 1$ for loss of function (LoF). A *residue* is a protein position of one gene; the substitutions at a residue share structural context and form one cluster. A gene's assayed missense variants are split by residue into a calibration part $\mathcal{A}$ and a certification part $\mathcal{B}$; the variants to be called (for example VUS) form the target. We write $q(s) \in [0, 1]$ for the score after Platt scaling [27] fitted to confidently labelled ClinVar variants (pathogenic or likely pathogenic, P/LP, against benign or likely benign, B/LB); $q(s)$ estimates the probability that such a ClinVar variant is pathogenic, not that an assay outcome is LoF. Because q is increasing in s, every procedure below makes the same calls on q as on s; q only puts predictors on one scale (the baselines use its values), so no ClinVar label affects a CertiVar call.

Per-class thresholds. The nonconformity of label 1 is $r_1 = 1 - q$ and of label 0 is $r_0 = q$. From each residue in $\mathcal{A}$ we keep one variant chosen uniformly at random. For class k, let n_k be the number of kept variants with label k and $r^{(k)}_{(1)} \le \cdots \le r^{(k)}_{(n_k)}$ their sorted nonconformities. The threshold is $\hat{t}_k = r^{(k)}_{(j_k)}$ (∞ if $j_k > n_k$), where j_k is the in-expectation rank or the probably approximately correct (PAC) rank ($j_k = n_k + 1$ if no j qualifies):

$$j_k = \lceil (n_k + 1)(1 - \alpha_k) \rceil \text{ (in expectation)} \quad \text{or} \quad j_k = \min\{\, j : \; F_{\mathrm{Beta}(j,\, n_k - j + 1)}(1 - \alpha_k) \le \delta \,\} \text{ (PAC).} \tag{1}$$

Algorithm 1 CertiVar for one gene and one evidence level

Require: scores q; assay labels split by residue into $\mathcal{A}, \mathcal{B}$; grid G; δ; targets τ (LoF) and τ' (functional)

1: keep one variant per residue of $\mathcal{A}$ and of $\mathcal{B}$
2: **for** $\alpha \in G$ **do**
3: thresholds $\hat{t}_0, \hat{t}_1$ on the kept variants of $\mathcal{A}$ by (1); call the kept variants of $\mathcal{B}$
4: $L^+(\alpha) \leftarrow \ell_{\mathrm{TP}}/u_{\mathrm{FP}}$, $U^-(\alpha) \leftarrow u_{\mathrm{FN}}/\ell_{\mathrm{TN}}$ (Clopper–Pearson, level $\delta/(2|G|)$ each)
5: **end for**
6: $\hat{\alpha}^+ \leftarrow \arg\max_{\alpha \in G}\{\text{share called LoF on } \mathcal{B} : L^+(\alpha) > \tau\}$
7: $\hat{\alpha}^- \leftarrow \arg\max_{\alpha \in G}\{\text{share called functional on } \mathcal{B} : U^-(\alpha) < \tau',\ \alpha \geq \hat{\alpha}^+\}$ (over all $\alpha \in G$ if $\hat{\alpha}^+$ does not exist)
8: **output** LoF calls at $\hat{\alpha}^+$ and functional calls at $\hat{\alpha}^-$ with their certified levels (no calls of a type whose set is empty); abstain otherwise

The prediction set is $C(q) = \{k : r_k \leq \hat{t}_k\}$; CertiVar calls LoF if $C = \{1\}$, functional if $C = \{0\}$, and abstains otherwise. A functional call on a LoF variant requires $1 \notin C$, so per-class coverage bounds both error rates.

Certified likelihood ratios. The ACMG/AMP guidelines [3] grade evidence in qualitative categories; in their Bayesian adaptation, supporting, moderate, strong and very strong evidence towards pathogenicity correspond to a likelihood ratio above 2.1, 4.3, 18.7 and 350, and supporting, moderate and strong evidence towards benignity to one below 0.48, 0.23 and 0.053 [4, 5]. We use these values as levels and write, for example, *LoF-supporting* for LoF calls whose likelihood ratio exceeds 2.1. Our likelihood ratios are defined with respect to the assay outcome Y and averaged over all variants that receive a call, as OddsPath is for a functional assay [5]; they are not the local, score-specific ratios that ClinGen uses for the computational criteria PP3/BP4 [2]. A level thus describes the call class as a whole and need not hold for each call (Section 5), and states how strongly a call predicts the assay result. It is evidence about pathogenicity only to the extent that the assay is clinically validated, and it must not be combined with the same assay's functional evidence (PS3/BS3) or with PP3/BP4 from the same predictor. For thresholds t, the call rates (probabilities, not counts) are $\mathrm{TP}(t) = \Pr(\text{call LoF} \mid Y = 1)$, $\mathrm{FP}(t) = \Pr(\text{call LoF} \mid Y = 0)$, $\mathrm{FN}(t) = \Pr(\text{call functional} \mid Y = 1)$ and $\mathrm{TN}(t) = \Pr(\text{call functional} \mid Y = 0)$, and $\mathrm{LR}^+ = \mathrm{TP}/\mathrm{FP}$, $\mathrm{LR}^- = \mathrm{FN}/\mathrm{TN}$.

For each α on a grid G we compute thresholds on $\mathcal{A}$ ($\alpha_0 = \alpha_1 = \alpha$), call one variant per residue of $\mathcal{B}$, and bound the four rates by one-sided Clopper–Pearson bounds [28] at level $\epsilon = \delta/(2|G|)$, written ℓ (lower endpoint) and u (upper endpoint) to keep them apart from the rates themselves: for x calls among n variants of a class, $\ell = F^{-1}_{\mathrm{Beta}(x,n-x+1)}(\epsilon)$ (0 if $x = 0$) and $u = F^{-1}_{\mathrm{Beta}(x+1,n-x)}(1-\epsilon)$ (1 if $x = n$, including $n = 0$). The certified bounds are $L^+(\alpha) = \ell_{\mathrm{TP}}/u_{\mathrm{FP}}$ and $U^-(\alpha) = u_{\mathrm{FN}}/\ell_{\mathrm{TN}}$ ($U^- = \infty$, never meeting a level, if $\ell_{\mathrm{TN}} = 0$; $u_{\mathrm{FP}} > 0$ always; for point estimates, $x/0 = \infty$ for $x > 0$, and $0/0$ is undefined and never selected). For each level, Algorithm 1 returns the α that calls the most variants among those whose certified bound meets the level. Because $\hat{t}_0$ and $\hat{t}_1$ both decrease weakly in α, no variant can be called LoF at one α and functional at another (Section 3.1). Line 7 also considers, for functional calls, only α at least as large as the one selected for LoF calls of the same strength; this is not needed for validity and can only lower the functional yield, so the yields we report are conservative. The *fixed-sequence* variant replaces the Bonferroni level $\delta/(2|G|)$ by $\delta/2$ and tests the grid in an order fixed in advance, stopping at the first α that does not meet the level (Theorem 2). We test LoF levels in increasing α and functional levels in decreasing α, because few variants are called functional at small α. Computing all thresholds and bounds costs $O(|G|\, n \log n)$ for n calibration and certification residues, about 5 ms per gene on a laptop CPU.

3.1 Guarantees

Let Q be the law of a variant obtained by drawing a residue from the gene's population of assayed residues and then one of its variants uniformly at random (binary-labelled, non-splice variants outside the ClinVar labelled set).

Assumption 1 (Clustered sampling)**.** Residues in $\mathcal{A}$, in $\mathcal{B}$ and of a test variant are independent draws from one residue population; within a residue, the kept or tested variant is chosen uniformly at random; q does

not depend on these data.

Assumption 2 (Class-conditional invariance)**.** Target variants have, for $k \in \{0, 1\}$, the same law of q given $Y = k$ as a test variant under Assumption 1; their LoF share $\pi = \Pr(Y = 1)$ may differ.

Proposition 1 (Per-class coverage)**.** *Under Assumption 1, for a fixed α_k, with the in-expectation rank and conditionally on n_k, $1 - \alpha_k \le \Pr(k \in C \mid Y = k, n_k) \le 1 - \alpha_k + 1/(n_k + 1)$, the upper bound holding when there are no ties.*

Proposition 2 (High-probability per-class coverage)**.** *Under Assumption 1, for a fixed α_k and with the PAC rank, $\Pr(k \in C \mid Y = k, \hat{t}_k) \ge 1 - \alpha_k$ with probability at least $1 - \delta$ over the calibration data for each k, and for both classes with probability at least $1 - 2\delta$.*

Proposition 3 (Coverage after selection)**.** *Under Assumption 1, for any $\hat{\alpha} \in G$ chosen from data independent of the test variant (for example by Algorithm 1), the in-expectation rank gives $\Pr(k \in C \mid Y = k) \ge 1 - \max G$; the PAC rank at level $\delta/|G|$ gives $\Pr(k \in C \mid Y = k, \hat{t}_k) \ge 1 - \hat{\alpha}$ for each k with probability at least $1 - \delta$.*

Theorem 1 (Certified likelihood ratios)**.** *Under Assumption 1, conditionally on $\mathcal{A}$ and with probability at least $1 - \delta$ over $\mathcal{B}$, $\mathrm{LR}^+(\hat{t}(\alpha)) \ge L^+(\alpha)$ under Q for every $\alpha \in G$ simultaneously; the same holds for $\mathrm{LR}^- \le U^-$ with a second δ. Because the event does not depend on the level, each call type returned by Algorithm 1 reaches its certified level at every evidence level reported, with probability at least $1 - \delta$ (both call types with probability at least $1 - 2\delta$), whichever α is selected.*

Theorem 2 (Fixed-sequence certification)**.** *Under Assumption 1, fix a level τ and an order of G chosen in advance. Reject $H_\alpha : \mathrm{LR}^+(\hat{t}(\alpha)) \le \tau$ when $\ell_{\mathrm{TP}}/u_{\mathrm{FP}} > \tau$ with both endpoints at level $\delta/2$, stop at the first α not rejected, and select the rejected α with the largest yield. Then, conditionally on $\mathcal{A}$, the selected LoF calls have $\mathrm{LR}^+ > \tau$ with probability at least $1 - \delta$. The same holds for functional calls with $u_{\mathrm{FN}}/\ell_{\mathrm{TN}} < \tau'$.*

Corollary 1 (No prior needed)**.** *Under Assumptions 1 and 2, Propositions 1–3 and Theorems 1–2 hold for target variants whatever their LoF share $\pi \in (0, 1)$: for a fixed α, LoF variants are called functional at rate at most α_1, and the certified likelihood ratios with respect to the assay outcome need no prior.*

Propositions 1 and 2 are the class-conditional and training-conditional conformal guarantees [12–15], here for residue-clustered data; Theorems 1 and 2 follow the risk-control and Learn-then-Test templates [19, 20] for a ratio of class-conditional call rates. Full proofs are in Appendix S1; we sketch three of them.
Proof sketches. Proposition 3: given $\mathcal{A}$, both ranks in (1) are non-increasing in α, so $\hat{t}_k(\hat{\alpha}) \ge \hat{t}_k(\max G)$ and coverage at $\hat{\alpha}$ is at least coverage at $\max G$; for the PAC rank, a union bound over the $|G|$ thresholds of class k gives the second claim. Theorem 1: given $\mathcal{A}$ the thresholds are fixed, and given the labels of the kept variants of $\mathcal{B}$ the class-k kept variants are independent draws from $Q(\cdot \mid Y = k)$, so each of the four counts is binomial with the corresponding call rate. Each of the $2|G|$ Clopper–Pearson endpoints entering L^+ fails with probability at most $\delta/(2|G|)$, so all hold with probability at least $1 - \delta$, and on that event $L^+(\alpha) \le \mathrm{LR}^+(\hat{t}(\alpha))$ for every $\alpha \in G$, including any selected one. Theorem 2: if $\mathrm{LR}^+ \le \tau$ and both endpoints hold, then $\ell_{\mathrm{TP}} \le \mathrm{TP} \le \tau\,\mathrm{FP} \le \tau\, u_{\mathrm{FP}}$, so a false rejection requires an endpoint to fail. A false claim requires rejecting the first true H_α in the order, which happens with probability at most δ. LoF and functional calls are disjoint for any $\alpha^+, \alpha^- \in G$ because $\hat{t}_0$ and $\hat{t}_1$ are both non-increasing in α (Appendix S1).

Scope. Propositions 1 and 2 hold for a fixed α. After Algorithm 1 selects $\hat{\alpha}$, the likelihood-ratio certificates still hold (Theorems 1 and 2), but with the in-expectation rank per-class coverage is guaranteed only at $1 - \max G$ (Proposition 3); we report the coverage achieved at the selected α in Section 4.4. Theorem 1 gives a numerical bound valid for all levels at once; Theorem 2 certifies one level at a time and does not give a numerical bound. Assumption 1 treats assayed residues as exchangeable, which a contiguous region need not be; Assumption 2 holds only approximately for VUS, on which AlphaMissense discriminates less well (Section 4.1), and not for variants whose assay result would be intermediate. Table 2 lists the experiments that test each guarantee.

Table 1: Datasets. *ClinVar labelled*: missense single-nucleotide variants (SNVs) with ≥2 review stars and a P/LP or B/LB call (used only to fit the common score scale) and the number that are P/LP. All shares are variant-level (descriptive). *Assayed*: missense SNVs with a binary assay label that are not in the ClinVar labelled set, and their LoF share; *assayed VUS*: the subset classified VUS in ClinVar. Assay labels are the authors' published classes (authors) or, where none are deposited, derived from the assay's own synonymous and nonsense controls (controls).

			ClinVar labelled		Assayed		Assayed VUS	
Gene	Assay	Labels	n	P/LP	n	LoF	n	LoF
BRCA1	SGE [29]	authors	345	156	1,617	0.161	272	0.173
BRCA2	SGE [30]	controls	309	94	3,435	0.130	704	0.207
BAP1	SGE [31]	authors	14	6	4,294	0.213	1,069	0.166
RAD51C	SGE [32]	authors	13	10	2,225	0.259	550	0.396
MSH2	DMS [33]	authors	178	121	2,906	0.060	1,542	0.067
Total			859	387	14,477		4,137	

4 Experiments

4.1 Data and protocol

We took every GRCh38 single-nucleotide missense variant (SNV) in BRCA1, BRCA2, BAP1, RAD51C and MSH2 that is in ClinVar [1] (release of 26 October 2026) or was assayed, annotated each on its Matched Annotation from NCBI and EMBL-EBI (MANE) Select transcript with the Ensembl Variant Effect Predictor [34] (AlphaMissense [35], REVEL [36]) and computed ESM-2 650M [37] log-likelihood ratios by wild-type marginals [38]. Labels come from five MaveDB [39] score sets (Table 1; accessions and versions in Appendix S5). For BRCA1, BAP1 and RAD51C we used the assay authors' classes (for BAP1 recomputed from the deposited false-discovery rates and score signs, which reproduces the deposited class counts exactly), and for MSH2 the authors' threshold (a positive loss-of-function score [33]). BRCA2 deposits no per-variant classes in MaveDB, so its labels come from Gaussians fitted to the assay's own synonymous and nonsense variants (LoF if the posterior is at least 0.95, functional if at most 0.05). Compared with the authors' classes, this control rule made no opposite call in BRCA1 or RAD51C, but it gave the opposite call for 216 of the 4,023 BAP1 variants that both rules label and left 77% of the authors' functional RAD51C variants intermediate, so we repeat the main experiments under alternative label rules (Section 4.5). We excluded the 729 assayed missense variants that also touch a splice region, since SGE also measures splicing, and set aside the 1,274 intermediate variants (Section 4.5). No ClinVar label enters an assay label.

One Platt map per predictor, pooled over genes, used the 859 ClinVar missense variants with at least two review stars and a P/LP or B/LB call (45.1% P/LP); these are excluded from every assay-label evaluation set, which leaves 14,477 assayed variants with a binary label. Of the 859, 422 also have a binary assay label, and for these P/LP matched LoF and B/LB matched functional for 97.4%, partly because some ClinVar classifications already use these assays. Pooled over genes, AlphaMissense has an area under the ROC curve (AUROC) of 0.946 on the ClinVar set, 0.840 on the other assayed variants and 0.779 on assayed VUS (non-overlapping 95% residue-bootstrap intervals), a drop seen for other predictors in DMS benchmarks [40]. Baselines are pooled (marginal) split conformal prediction on the same calibration data; label-shift-weighted conformal prediction [26] with the true target share or its expectation–maximization (EM) estimate [24]; fixed tiers ($q \geq 0.9$ LoF, $q \leq 0.1$ functional); the published AlphaMissense classes (likely pathogenic > 0.564, likely benign < 0.34); and the REVEL PP3/BP4 supporting thresholds (≥ 0.644, ≤ 0.290) [2]. None claims per-class coverage, and the published cut-offs target pathogenicity, so we judge them only as assay classifiers. Coverage is residue-weighted as in Q; *worst-class coverage* is the smaller of the mean LoF and mean functional coverages over repetitions. Table 2 lists the designs.

Table 2: Experimental designs. Every split is by residue; calibration keeps one variant per residue.

Experiment	Residue split	Tests	Reps
Coverage (4.2)	halves: calibrate on one, test on the other	Prop. 1	50
Calibration share (4.2)	halves; calibration half subsampled to LoF share 0.05–0.5	Cor. 1	100
Reliability (4.3)	halves	Props. 1–2	200
VUS (4.3)	halves; non-VUS of one half (40% calibrate, 60% certify); VUS of the other	Cor. 1	50
Oracle (4.4)	30% and 50% of residues drawn with replacement; truth on all residues	Thms. 1–2, Prop. 3	100
Held-out fifth (4.4)	disjoint 30% / 50% / 20%	Thm. 1	100
Regions (4.5)	three contiguous quarters / the fourth	beyond Asm. 1	25/quarter

4.2 Per-class coverage, and a shift in the calibration LoF share

In each gene we split the residues in half 50 times, calibrated on one half and evaluated on the other. At $\alpha = 0.1$ the in-expectation rule gave worst-class coverage between 0.896 and 0.908 in all 15 gene–predictor pairs, none more than one Monte Carlo standard error (0.002–0.010) below the target (Table 3, Figure 2A); it was at least 0.948 at $\alpha = 0.05$ and at least 0.792 at $\alpha = 0.2$. Pooled conformal fell to 0.545 on LoF variants in its worst pair and below 0.85 in 5 pairs, because a marginal threshold spends its error budget on the rarer class; where it covered LoF variants, as for AlphaMissense in BAP1, it abstained on 69% of variants. The fixed tiers fell below 0.85 in 5 pairs, and the published AlphaMissense and REVEL cut-offs covered functional variants as little as 0.540 (BAP1) and 0.445 (MSH2). The per-class rule abstained on 35% of variants on average (17–47%), against 42% for pooled conformal, 53% for the fixed tiers, 42% for the REVEL cut-offs and 11% for the AlphaMissense classes, which trade coverage for decisiveness. In a separate set of 50 splits, calibrating on all variants of the calibration half instead of one per residue used five times as many labels (1,446 against 292 on average) and lowered abstention from 35.2% to 30.4%, but worst-class coverage fell below 0.90 in 13 of 15 pairs (0.887–0.903) against 4 of 15 (0.897–0.915), as expected when substitutions at one residue are dependent.

To test the claim that per-class thresholds need no prevalence estimate, we subsampled each calibration half (one variant per residue, 36 to 176 variants depending on the gene) to a LoF share of 0.05, 0.1, 0.2, 0.3 or 0.5 and applied the thresholds to the other half at its natural share (100 repetitions; Figure 4A). The per-class rule kept worst-class coverage between 0.897 and 0.925 in every pair at every share (0.891–0.925 in a separate run that balanced the whole calibration half to 50% LoF), at the cost of abstention that rose from 37% at share 0.5 to 83% at share 0.05, when few LoF variants remain to set the LoF threshold. Pooled conformal calibrated at share 0.05 covered LoF variants at 0.113 in its worst pair, and label-shift-weighted conformal fell to 0.53–0.60 in its worst pair with the true target share and to 0.04 with the EM estimate. The estimate is the weak point: from unlabelled ClinVar-calibrated scores, EM missed the LoF share of a gene's assayed variants by 0.255 on average and by up to 0.702 (black-box shift estimation [25]: 0.233; Appendix S3), because ClinVar pathogenicity and assay loss of function differ in label definition as well as in class share.

4.3 Reliability across splits and a real change of population

With 17 to 93 LoF residues per calibration half, a single split can fall short of the in-expectation guarantee: over 200 splits per pair, LoF coverage was below 0.90 in 36.4% of splits and functional coverage in 43.6% (Figure 2B). PAC thresholds ($\delta = 0.1$) reduced these to 7.9% and 10.8% (the latter above δ because coverage on a finite test half is a noisy estimate of the true coverage) at the price of abstaining on 58.4% of variants against 35.4%. To test a real change of population, we set thresholds on non-VUS variants of one half of each gene's residues and called the assayed VUS of the other half (mean residue-weighted LoF share 17.0% in calibration and 19.8% among VUS; 50 repetitions). In-expectation thresholds kept LoF coverage between 0.854 and 0.958 and functional coverage between 0.841 and 0.942, each below 0.90 in 7 of 15 pairs (Figure 2C). PAC thresholds gave LoF coverage between 0.957 and 1.000 and functional coverage between

Table 3: Calls by gene and predictor ($\alpha = 0.1$; calibration and test on disjoint halves of the gene's residues, mean of 50 splits). Per-class rule: residue-weighted coverage of LoF and functional (Fun) variants and abstention rate (Abs); Monte Carlo standard errors of the coverages are 0.002–0.010. Baselines: worst-class coverage (bold if below 0.85) and abstention. Published cut-offs: AlphaMissense classes (AM) and REVEL PP3/BP4 thresholds; none exist for ESM-2.

		Per-class (ours)			Pooled conformal		Published cut-off	
Gene	Predictor	LoF	Fun	Abs	Worst	Abs	Worst	Abs
BRCA1	AM	0.916	0.901	0.25	0.889	0.32	**0.773**	0.13
	REVEL	0.914	0.902	0.34	0.883	0.48	**0.663**	0.59
	ESM-2	0.911	0.897	0.37	0.893	0.34	–	–
BRCA2	AM	0.899	0.903	0.36	**0.808**	0.17	**0.810**	0.11
	REVEL	0.912	0.899	0.42	**0.841**	0.24	**0.803**	0.57
	ESM-2	0.896	0.903	0.36	0.894	0.51	–	–
BAP1	AM	0.909	0.898	0.39	0.880	0.69	**0.540**	0.07
	REVEL	0.903	0.904	0.42	**0.545**	0.15	**0.799**	0.29
	ESM-2	0.910	0.905	0.47	0.878	0.36	–	–
RAD51C	AM	0.901	0.905	0.17	**0.824**	0.10	0.860	0.11
	REVEL	0.909	0.904	0.27	**0.625**	0.20	**0.792**	0.25
	ESM-2	0.915	0.908	0.34	0.886	0.50	–	–
MSH2	AM	0.915	0.901	0.29	0.897	0.70	**0.642**	0.12
	REVEL	0.911	0.901	0.37	0.896	0.79	**0.445**	0.41
	ESM-2	0.920	0.902	0.46	0.895	0.80	–	–

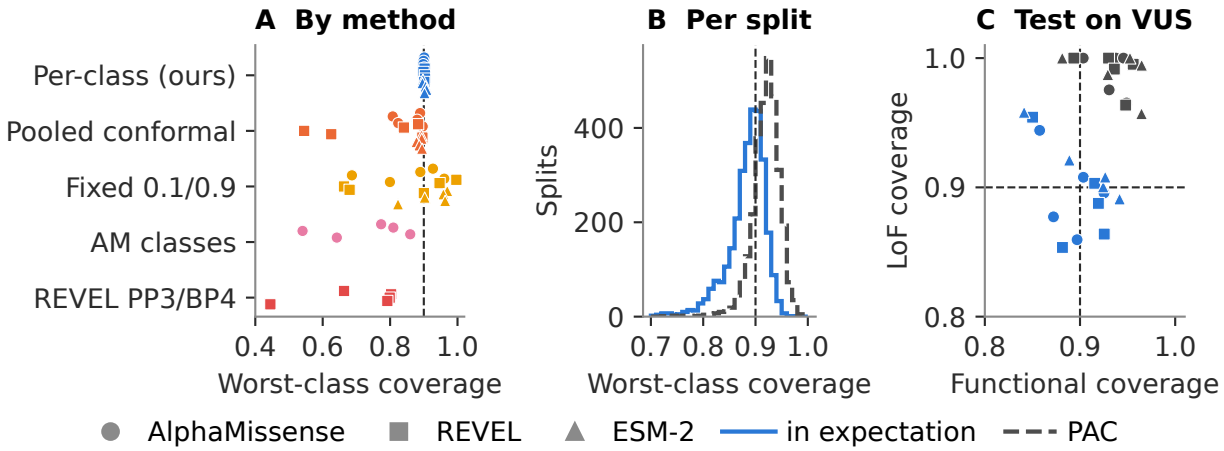


Figure 2: Per-class coverage at $\alpha = 0.1$. Markers give the predictor in panels A and C. (A) Worst-class coverage for each gene–predictor pair (smaller of the two per-class means over 50 splits). (B) Smaller of the LoF and functional coverages in each split, pooled over the 15 pairs (3,000 splits per rule). (C) LoF and functional coverage on assayed VUS with thresholds set on non-VUS variants at other residues (mean of 50 repetitions per pair).

0.881 and 0.965, below 0.90 in 2 pairs, but abstained on 81.7% of VUS against 46.4%, because the PAC rank is conservative when each class has few calibration residues.

4.4 Certified likelihood ratios

Treating each gene's assayed residues as the population, we drew calibration (30% of residues) and certification (50%) samples with replacement, 100 times per pair, ran Algorithm 1 with $G_4 = \{0.05, 0.10, 0.20, 0.30\}$, $\delta = 0.05$ and the in-expectation rank for each level, and computed the true residue-weighted likelihood

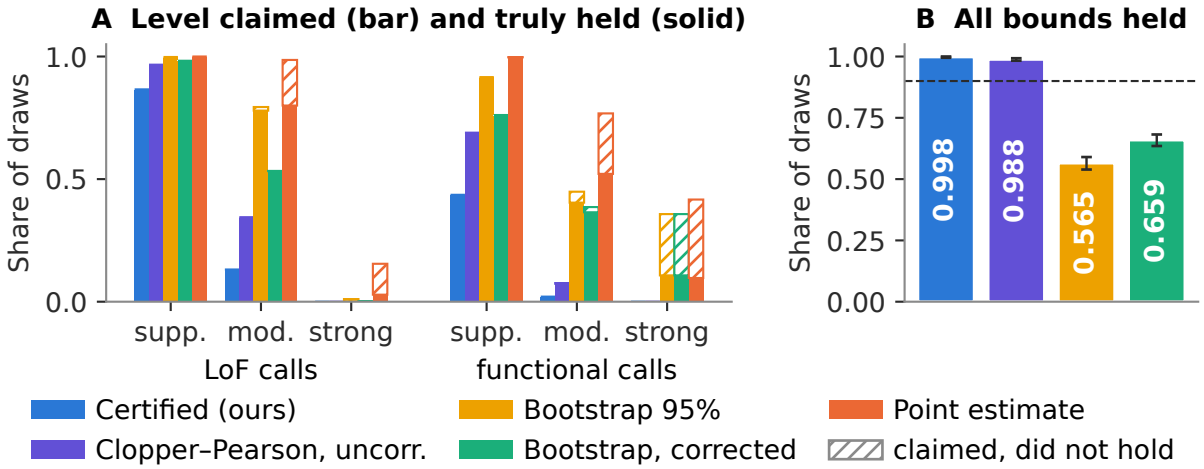


Figure 3: Selecting thresholds for a level, judged against the true likelihood ratio (oracle design, 100 draws per pair, 1,500 in all). (A) Share of draws in which each rule claims the level (full bar) and in which it truly held (solid part; hatched: claimed but did not hold); supp., mod. = supporting, moderate. (B) For the four rules that produce bounds, share of draws in which every claimed bound was respected (true $\mathrm{LR}^+ \geq L^+$, true $\mathrm{LR}^- \leq U^-$), with 95% intervals from resampling draws; dashed line at the guaranteed 0.90.

ratio of every selected rule on the whole population. We compared four other rules applied to the same data: Clopper–Pearson bounds without multiplicity correction, a one-sided 95% percentile bootstrap of the one-variant-per-residue certification sample (2,000 resamples), the same bootstrap at level $\delta/(2|G|)$, and the point estimate. Calibration workflows use the bootstrap for a local likelihood ratio [2], so this compares uncertainty estimates, not published thresholds. A bound is respected when the true LR^+ is at least L^+ (LoF calls) or the true LR^- is at most U^- (functional calls).

Every bound claimed in a draw was respected in 99.8% of the 1,500 draws for the certified rule (Theorem 1 guarantees at least 90%), 98.8% for uncorrected Clopper–Pearson bounds, 56.5% for bootstrap bounds and 65.9% for the corrected bootstrap (Figure 3B). Per selection, the certified bound was respected in 2,178 of 2,181 cases (99.9%), uncorrected bounds in 99.2% of 3,115 and bootstrap bounds in 76.5% of 5,287. Most bootstrap failures are degenerate resamples: in a fresh set of 1,500 draws, 57% of failed bootstrap bounds came from certification samples with no false call of that type, where the bound failed 89% of the time, but bounds failed in 11% of cases even with nonzero counts. A point estimate is not a bound; judged by whether the claimed level truly held, the certified rule held in all 2,181 claims, bootstrap bounds in 91.0% and point estimates in 79.7%: point estimates claimed LoF-moderate in 98.6% of draws and it held in 81.1% of claims (Figure 3A). The true ratio weighted by variants instead of residues gave the same conclusions (levels held in 99.9% of certified claims). On the four-level grid, the exact binomial bound on one variant per residue, not the multiplicity correction, separates the valid rules from the others; the correction costs yield (LoF-moderate certified in 13.1% of draws with it, 34.5% without).

CertiVar certified LoF-supporting calls in 86.6% of draws (74–97% by gene; 25.8% of variants called), functional-supporting calls in 43.6% (61.1% called), LoF-moderate in 13.1% and functional-moderate in 2.1%, and never a strong level. In a fresh set of 1,500 draws, at the selected α (mean 0.18 for LoF and 0.24 for functional calls), per-class coverage on the population averaged 0.82 for LoF and 0.80 for functional variants, above the 0.70 that Proposition 3 guarantees for $\max G_4 = 0.3$, but it fell below 0.69 in a tenth of selections: selecting for evidence strength trades coverage for confidence in the calls that are made. On the same 1,500 fresh draws, fixed-sequence testing (Theorem 2) raised the share of draws certifying LoF-moderate from 12.9% to 29.1% and functional-supporting from 42.5% to 60.6%, with claimed levels holding in 99.7% of 2,705 claims and no level falsely claimed in more than 0.4% of draws (Figure 4B). On a 20-level grid G_{20} (α from 0.02 to 0.40) the Bonferroni level shrinks and LoF-moderate fell to 6.2%, whereas fixed-sequence testing kept 26.7%; it lost LoF-supporting yield (65.3% against 73.8%) because its first, smallest α calls few variants. Testing functional levels in increasing α instead certified functional-supporting in only 25.7% (G_4) and 0.2% (G_{20}) of draws, so the order matters and must be fixed in advance.

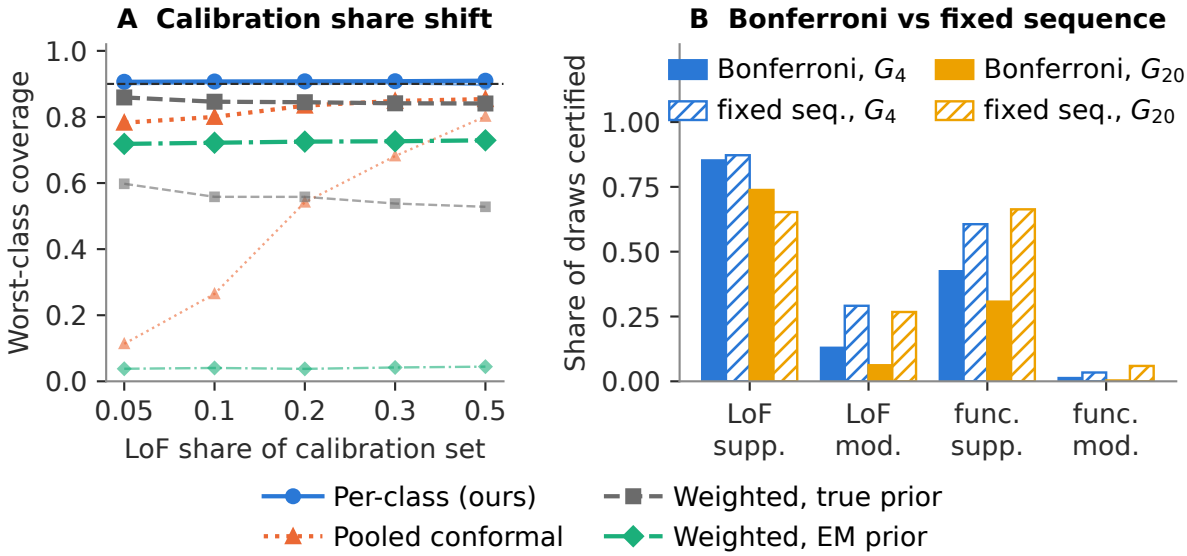


Figure 4: (A) Worst-class coverage when the calibration half is subsampled to a given LoF share and tested at its natural share: mean over the 15 pairs (thick) and worst pair (thin; for the per-class rule the two nearly coincide); 100 repetitions per pair. (B) Share of oracle draws (1,500) certifying each level with a Bonferroni correction (Algorithm 1) or fixed-sequence testing (Theorem 2), on the four-level grid G_4 and the 20-level grid G_{20}.

In data splits that held out a test fifth of the residues (100 per pair), the empirical residue-weighted likelihood ratio on the test fifth respected the certified bounds in 98.7% of 2,091 cases, against 76.8% for bootstrap bounds. Calibrated and certified on non-VUS variants and applied to VUS at other residues, the LoF-supporting level was certified in 59.1% of repetitions and the empirical likelihood ratio on the VUS met it in all 443; the functional-supporting level held in 123 of 125 and the LoF-moderate level in 25 of 28 (89%; 95% interval 72–98%).

The certificates concern the assay outcome. As an exploratory check against clinical labels, we set per-class thresholds ($\alpha = 0.1$) on all assay-labelled residues of each gene and called one ClinVar variant per residue among the 859 with at least two stars, which never enter the thresholds; likelihood-ratio bounds against P/LP versus B/LB use one-sided Clopper–Pearson endpoints at level 0.025 each (Table S4). In the three genes with enough labelled residues (BRCA1: 75 P/LP and 170 B/LB; BRCA2: 60 and 210; MSH2: 72 and 55), LoF calls had lower bounds of 4.9–19.6 in BRCA1, 7.6–9.4 in BRCA2 and 2.6–3.2 in MSH2 across predictors, and functional calls upper bounds of 0.24–0.37. Confidently classified variants are easier than typical VUS, and some ClinVar classifications already use these assays or predictors as evidence, so this is not independent clinical validation.

4.5 How much data, from where, and with which labels

In our five genes, calibration data had to come from the gene itself: calibrating on the other four genes left 8 of 15 pairs below 0.85, with a worst value of 0.487 (Fig. S2C), consistent with gene-specific evidence strength [10, 11]. Calibrating on three contiguous quarters of a protein and testing on the fourth gave worst-class coverage between 0.860 and 0.923; certified bounds were respected in the held-out quarter in 88.8% of 1,891 cases and the claimed level held in 93.0%, below the nominal 95%, because another region's residues are not exchangeable with the calibrated ones. With 150 calibration residues drawn at random, coverage averaged over genes and predictors met the target with 40.5% abstention, against 39.8% with 200 and 82.8% with 25 (Fig. S2A). Coverage is marginal within a class: on the third of variants whose assay score lies nearest the boundary between the two class medians it was 0.865 for LoF and 0.873 for functional variants (one split per pair, variant-weighted; Appendix S4). The 1,262 intermediate variants outside the ClinVar labelled set (BRCA2 888, BAP1 210, BRCA1 155, RAD51C 9), which no guarantee covers, were called LoF in 19.6% and functional in 36.6% of cases. Applied outside the assayed region, to 1,534 unassayed BRCA1 VUS,

AlphaMissense thresholds called 1,463 functional and abstained on only 53, far fewer abstentions than on assayed variants; this suggests a shifted score distribution there, which no guarantee covers. Finally, we relabelled BRCA2 with control-rule cut-offs of 0.80 and 0.99, and BAP1 and RAD51C with the control rule instead of the authors' classes (Table S3, an independent re-run with its own random splits). The label rule changed the yield: under the control rule 10% of labelled BAP1 variants were LoF rather than 21%, and abstention fell from 39–47% to 13–19%. It did not change validity: under the four alternative rules, worst-class coverage stayed between 0.894 and 0.908 across gene–predictor pairs, and certified bounds held in 2,530 of 2,533 oracle selections.

5 Discussion

In our resampling design, thresholds selected on point estimates or percentile-bootstrap bounds often claimed more evidence than held, and most bootstrap failures came from certification samples with no false call. Because these are call-averaged ratios, not the local ratios of ClinGen calibrations, this does not show that published PP3/BP4 thresholds are wrong, only that point estimates and bootstrap bounds can overstate evidence at single-gene sample sizes. CertiVar makes the claim checkable without an estimate of the share of damaging variants, and per-class coverage stayed at or near the target when the calibration LoF share was moved from 5% to 50%. The intended use is a gene with a functional assay: calling the variants that the assay missed within the assayed regions, and checking whether a predictor's claimed evidence strength holds against the assay. The cost is conservatism: certified evidence was mostly supporting-level, and PAC thresholds abstained on 58% to 82% of variants, suited to settings where overstating strength is worse than abstaining. Held-out protein regions, a proxy for the unassayed regions where predictors are most needed, fall outside the theory (bounds held in 88.8% of cases), so extrapolation needs local validation or a region-aware procedure.

Limitations.

- *Assay, not pathogenicity.* The labels are assay classes [5]; clinical use of the certified ratios requires an assay validated against clinical classifications, as its authors did for BRCA1 [29] and, with their own classes rather than our control-based labels, for BRCA2 [30].
- *Call-averaged, not local.* A certified level describes the calls as a class. Variants near the assay class boundary are covered less often (0.865 and 0.873 on the nearest third), so a single borderline call can carry less evidence than the level of its class; ClinGen uses local ratios for this reason [2].
- *Who is covered.* The guarantees assume residues exchangeable with the calibrated ones and do not cover intermediate or splice-region variants, which a clinician cannot exclude in advance; 19.6% of intermediate variants were called LoF. Per-class coverage holds for a fixed α; at the selected α only $1 - \max G$ is guaranteed.
- *Data.* One variant per residue discards most labels, but using all of them pushed coverage below the target in 13 of 15 pairs. Results cover five cancer-predisposition genes, four of them in DNA repair; other genes may need more calibration residues. REVEL was trained on labelled variants and the AlphaMissense classes were set on ClinVar, so both may overlap our evaluation sets [35, 36, 41]. ESM-2, trained without variant labels, gave the same coverage and validity but certified LoF-supporting calls less often (78% of oracle draws, against 86% for REVEL and 96% for AlphaMissense), so such overlap could inflate yields rather than validity.

6 Conclusion

CertiVar turns a variant-effect predictor and a few hundred assayed residues of a gene into LoF, functional or no calls with finite-sample guarantees on per-class error rates at a fixed miscoverage level and on call-averaged likelihood ratios with respect to the assay after threshold selection, without an estimate of how common damaging variants are. Empirically, the guarantees held within genes and under shifts of the calibration LoF share, approximately on assayed VUS, and less well in held-out protein regions. Clinical use requires clinically validated assays. Code and data: `https://github.com/aaryavvatts-lab/certivar` (release v1.0.0).

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Appendix

Section, figure and table numbers without the prefix S refer to the main paper. Contents: S1 proofs of Propositions 1–3, Theorems 1–2 and Corollary 1; S2 simulation (Fig. S1) and why marginal thresholds do not control the minority class; S3 calibration size, region and gene transfer (Fig. S2A–C), per-gene coverage (Table S1) and prior estimation (Table S2); S4 strata, assay-label sensitivity (Table S3) and calls judged against ClinVar labels (Table S4); S5 implementation.

S1 Proofs

Assumptions, propositions, theorems and the corollary are stated in Section 3.1; Q, C, $\hat{t}_k$, L^+ and U^- are defined in Section 3.

Throughout, fix a class k and write F_k for the distribution function of the nonconformity r_k of a variant drawn from $Q(\cdot \mid Y = k)$. By Assumption 1, the kept calibration variants (one per residue of $\mathcal{A}$) and the variant to be called are independent draws from Q; conditionally on all labels, the n_k kept variants of class k and a class-k test variant are independent draws from $Q(\cdot \mid Y = k)$.

Proposition 1. Conditionally on the labels, the $n_k + 1$ nonconformities of the class-k kept variants and of the test variant are exchangeable, so with ties broken at random the rank of the test value is uniform on $\{1, \ldots, n_k + 1\}$ [12, 42]. The test value is at most the j_k-th smallest calibration value with probability $j_k/(n_k + 1)$, which lies in $[1 - \alpha_k,\ 1 - \alpha_k + 1/(n_k + 1))$ for $j_k = \lceil (n_k + 1)(1 - \alpha_k) \rceil$; ties only increase the probability, which gives the lower bound in general. If $j_k > n_k$ then $\hat{t}_k = \infty$. Since $k \in C$ exactly when $r_k \le \hat{t}_k$, the claim follows. □

Proposition 2. Conditionally on the labels, the coverage of the threshold is $F_k(\hat{t}_k) = F_k(r^{(k)}_{(j)})$. If F_k is continuous, $F_k(r^{(k)}_{(j)})$ is the j-th order statistic of n_k independent uniforms and has the $\mathrm{Beta}(j, n_k - j + 1)$ distribution, so $\Pr\{F_k(\hat{t}_k) < 1 - \alpha_k\} = F_{\mathrm{Beta}(j, n_k - j + 1)}(1 - \alpha_k) \le \delta$ by the choice of j_k [14]. If F_k has atoms, write $r = F_k^{-1}(U)$ with U uniform and F_k^{-1} the generalized inverse; then $r^{(k)}_{(j)} = F_k^{-1}(U_{(j)})$ and $F_k(F_k^{-1}(u)) \ge u$, so $F_k(r^{(k)}_{(j)}) \ge U_{(j)}$ and the bound still holds. If no $j \le n_k$ qualifies, $\hat{t}_k = \infty$ and coverage is 1. □

Proposition 3. Fix $\mathcal{A}$. Both ranks in (1) are non-increasing in α (the in-expectation rank because $\lceil (n_k + 1)(1 - \alpha) \rceil$ is, the PAC rank because $F_{\mathrm{Beta}(j, n_k - j + 1)}(1 - \alpha)$ decreases in α for each j, so the smallest qualifying j cannot increase), so $\hat{t}_k(\alpha)$ is non-increasing in α and the set $\{r_k \le \hat{t}_k(\alpha)\}$ shrinks as α grows. Hence, for any $\hat{\alpha} \in G$ that depends on $\mathcal{B}$ but not on the test variant, $\Pr(k \in C_{\hat{\alpha}} \mid Y = k, \mathcal{A}, \mathcal{B}) \ge \Pr(k \in C_{\max G} \mid Y = k, \mathcal{A})$, and taking expectations over $\mathcal{A}$ and Proposition 1 at $\alpha_k = \max G$ give the first claim. For the PAC rank at level $\delta/|G|$, Proposition 2 applied to each $\alpha \in G$ and a union bound give, with probability at least $1 - \delta$, coverage at least $1 - \alpha$ for every $\alpha \in G$ simultaneously, in particular for $\hat{\alpha}$. □

Theorem 1. The thresholds $\hat{t}(\alpha)$ depend only on $\mathcal{A}$ and are fixed given $\mathcal{A}$. Conditionally on $\mathcal{A}$ and on the labels of the kept variants of $\mathcal{B}$, the class-1 kept variants of $\mathcal{B}$ are n_1 independent draws from $Q(\cdot \mid Y = 1)$, so the number of them called LoF is $\mathrm{Binomial}(n_1, \mathrm{TP}(\alpha))$ with $\mathrm{TP}(\alpha) = \Pr(\text{call LoF} \mid Y = 1)$, and likewise for the other three counts. Each one-sided Clopper–Pearson bound [28] at level $\delta/(2|G|)$ fails with probability at most $\delta/(2|G|)$; the conventions for $x = 0$, $x = n$ and $n = 0$ in Section 3 give the trivial bounds 0 and 1, which never fail. For LR^+ there are $2|G|$ such bounds (ℓ_{TP} and u_{FP} for each α), so by the union bound all of them hold simultaneously with probability at least $1 - \delta$. On that event $\mathrm{TP}(\alpha) \ge \ell_{\mathrm{TP}}(\alpha)$ and $\mathrm{FP}(\alpha) \le u_{\mathrm{FP}}(\alpha)$ for every $\alpha \in G$, hence $\mathrm{LR}^+(\hat{t}(\alpha)) \ge L^+(\alpha)$ for every α (with $\mathrm{LR}^+ = \infty$ when $\mathrm{FP}(\alpha) = 0$), including any $\hat{\alpha}^+$ chosen from these data. Since this probability bound holds for every $\mathcal{A}$, it also holds unconditionally. The argument for LR^- is identical with u_{FN} and ℓ_{TN}. The event does not involve the levels τ, so it covers every evidence level reported. □

Theorem 2. Fix $\mathcal{A}$, so the thresholds and the hypotheses $H_\alpha : \mathrm{LR}^+(\hat{t}(\alpha)) \le \tau$ are fixed. If H_α is true and both endpoints $\ell_{\mathrm{TP}}(\alpha)$ and $u_{\mathrm{FP}}(\alpha)$ at level $\delta/2$ hold, then $\ell_{\mathrm{TP}} \le \mathrm{TP} \le \tau\,\mathrm{FP} \le \tau\, u_{\mathrm{FP}}$, so H_α is not rejected; hence a true H_α is rejected with probability at most δ. Let α^* be the first α in the fixed order for which H_α is true (if none, every claim is true). The procedure claims the level at a false α only if it rejects every hypothesis up to that α, which includes H_{α^*}; this has probability at most δ. The selected α is one of the rejected ones, so the claimed level holds with probability at least $1-\delta$. The functional case is identical with $u_{\mathrm{FN}}/\ell_{\mathrm{TN}} < \tau'$. □

Corollary 1. Every quantity bounded above is a probability conditional on the true class: $\Pr(k \in C \mid Y = k)$, the four call rates, and their ratios. Under Assumption 2 these are the same for target variants as for a test variant under Assumption 1, whatever π is, so Propositions 1–3 and Theorems 1–2 transfer. The two call sets are disjoint for any $\alpha^+, \alpha^- \in G$: a LoF call at α^+ requires $q > \hat{t}_0(\alpha^+)$ and $1-q \le \hat{t}_1(\alpha^+)$, a functional call at α^- requires $q \le \hat{t}_0(\alpha^-)$ and $1-q > \hat{t}_1(\alpha^-)$, and both thresholds are non-increasing in α, so the first conditions contradict each other if $\alpha^- \ge \alpha^+$ and the second if $\alpha^- < \alpha^+$. A functional call on a LoF variant requires $1 \notin C$, so its rate is at most $1 - \Pr(1 \in C \mid Y = 1) \le \alpha_1$. None of these quantities involves π, so no prior is needed to certify them; this concerns the likelihood ratio with respect to the assay outcome, not evidence about pathogenicity. □

S2 Simulation

Scores were drawn per residue as $s = 2y + u_c + e$, with a residue effect $u_c \sim N(0, 0.5^2)$ shared by the five variants of a residue, unit Gaussian noise, and the class shared within a residue. A Platt calibrator was fitted on 600 variants at LoF share 0.45, conformal thresholds were set on another 600 (one variant per residue), and 3,000 test variants were drawn at LoF shares from 0.05 to 0.60, with 300 replicates per share (Figure S1). Per-class thresholds kept LoF coverage at the target at every share; pooled thresholds did not, and reweighting them to the EM estimate of the share [26] kept marginal coverage near the target while LoF coverage fell further as LoF variants became rarer. Adjusting posteriors to the EM estimate removed the calibration error caused by the shift in this simulation, where the shift is a pure change of prior, unlike in the real data (Section 4.2).

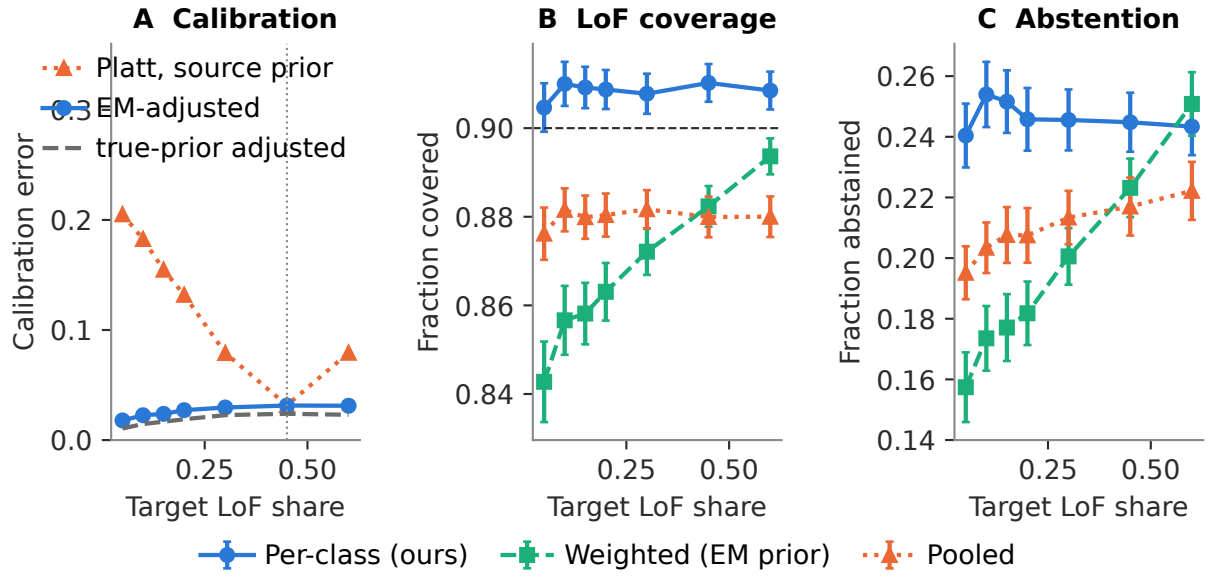


Figure S1: Simulation with residue-clustered scores, calibration at LoF share 0.45 and test share on the horizontal axis (300 replicates; error bars are 95% intervals of the mean). (A) Expected calibration error of Platt scaling at the source share, after EM prior adjustment, and after adjustment to the true share. (B) Coverage of LoF variants. (C) Abstention rate.

Why marginal thresholds do not control the minority class. A pooled split-conformal threshold $\hat{t}$ is a quantile of the mixture $\pi_c F_1 + (1-\pi_c)F_0$ of the two class-conditional nonconformity laws, weighted by the calibration share π_c. Its coverage of class 1 is $F_1(\hat{t})$, which is not controlled; when LoF variants are rare and harder to separate, $F_1(\hat{t}) < 1-\alpha$ while marginal coverage is still near $1-\alpha$. Reweighting to a known test prior [26] restores marginal coverage at that prior but does not control $F_1(\hat{t})$ either.

S3 Calibration size, region and gene transfer, and per-gene results

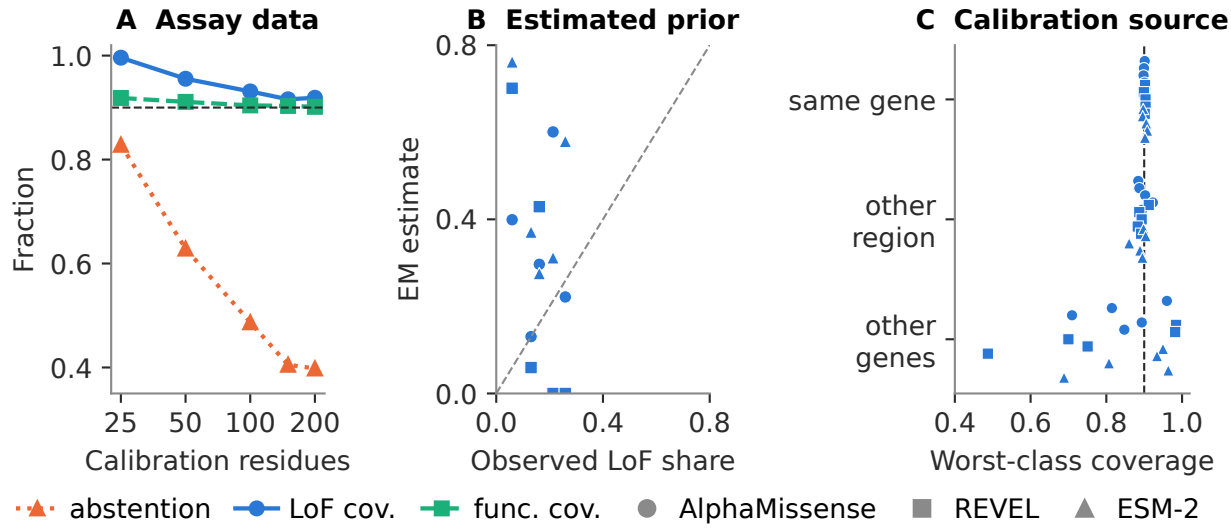


Figure S2: (A) Coverage and abstention against the number of calibration residues drawn from the gene (mean over genes, predictors and 50 draws). (B) EM estimate of the LoF share from unlabelled scores against the observed share; the dashed line is equality. (C) Worst-class coverage when calibrating on random residues of the same gene, on the other three quarters of the protein, or on the other four genes.

Table S1 gives per gene and predictor the residue-weighted coverage of each class and the abstention rate of the per-class rule and two baselines; Table S2 gives the estimated LoF shares and calibration errors behind Section 4.2. When the calibration and test halves of a gene share their LoF share, label-shift-weighted conformal prediction [26] with the true share nearly coincides with pooled conformal; with the EM estimate it inherits the estimation error, which collapsed its LoF coverage in some pairs (worst-class coverage as low as 0.039 at $\alpha = 0.1$).

S4 Strata, assay-label sensitivity and ClinVar check

For one within-gene split per gene and predictor ($\alpha = 0.1$, variant-weighted), coverage of LoF and functional variants was 0.966 and 0.929 on the third of variants farthest from the assay's class boundary, 0.929 and 0.912 on the middle third, 0.865 and 0.873 on the nearest third, and 0.881 and 0.896 on variants classified VUS in ClinVar, with abstention rates of 0.334, 0.385, 0.413 and 0.392.

Table S3 gives, for each label rule, the number of labelled variants, their LoF share, the coverage and abstention of the per-class rule, and the oracle check of the certified selection (Section 4.5). This analysis is an independent re-run with its own random splits, so its main-label rows differ from Table 3 by Monte Carlo error (for example 0.895 against 0.896 in the lowest pair).

Table S4 gives the exploratory check of the calls against ClinVar labels (Section 4.4), per gene and pooled. Only BRCA1, BRCA2 and MSH2 have enough labelled residues for informative bounds; across them the LoF-call lower bounds differ by gene, so the pooled values should not be read as a common likelihood ratio. Some ClinVar classifications already use these assays or predictors as evidence, so this check is not independent clinical validation.

Table S1: Within-gene random split, $\alpha = 0.1$, mean of 50 splits. Each cell is residue-weighted LoF coverage / functional coverage / abstention rate.

Predictor	Gene	Per-class (ours)	Pooled conformal	Fixed 0.1/0.9
AlphaMissense	BRCA1	0.916/0.901/0.25	0.952/0.889/0.32	0.971/0.927/0.44
AlphaMissense	BRCA2	0.899/0.903/0.36	0.808/0.921/0.17	0.890/0.965/0.37
AlphaMissense	BAP1	0.909/0.898/0.39	0.977/0.880/0.69	0.942/0.687/0.26
AlphaMissense	RAD51C	0.901/0.905/0.17	0.824/0.929/0.10	0.961/0.973/0.37
AlphaMissense	MSH2	0.915/0.901/0.29	0.972/0.897/0.70	0.956/0.799/0.39
REVEL	BRCA1	0.914/0.902/0.34	0.962/0.883/0.48	0.997/0.999/0.98
REVEL	BRCA2	0.912/0.899/0.42	0.841/0.911/0.24	0.947/0.998/0.68
REVEL	BAP1	0.903/0.904/0.42	0.545/0.997/0.15	0.663/1.000/0.24
REVEL	RAD51C	0.909/0.904/0.27	0.625/0.999/0.20	0.680/1.000/0.23
REVEL	MSH2	0.911/0.901/0.37	0.991/0.896/0.79	0.991/0.900/0.80
ESM-2	BRCA1	0.911/0.897/0.37	0.912/0.893/0.34	0.970/0.975/0.64
ESM-2	BRCA2	0.896/0.903/0.36	0.944/0.894/0.51	0.964/0.958/0.71
ESM-2	BAP1	0.910/0.905/0.47	0.878/0.912/0.36	0.903/0.943/0.46
ESM-2	RAD51C	0.915/0.908/0.34	0.973/0.886/0.50	0.987/0.963/0.71
ESM-2	MSH2	0.920/0.902/0.46	1.000/0.895/0.80	0.993/0.823/0.68

Table S2: Prior estimation and calibration on assayed variants outside the ClinVar labelled set. LoF share observed and estimated from unlabelled scores by expectation–maximization (EM) [24] and black-box shift estimation (BBSE) [25]; expected calibration error (10 equal-mass bins) of the ClinVar-Platt posterior, after EM prior adjustment, after adjustment to the true prior, and of Platt scaling fitted on the other four genes' assay labels. AM: AlphaMissense.

		LoF share			Expected calibration error			
Pred.	Gene	obs.	EM	BBSE	ClinVar Platt	+ EM	+ true prior	other genes
AM	BRCA1	0.161	0.297	0.316	0.203	0.135	0.064	0.049
AM	BRCA2	0.130	0.131	0.178	0.132	0.032	0.032	0.041
AM	BAP1	0.213	0.601	0.588	0.341	0.388	0.255	0.119
AM	RAD51C	0.259	0.222	0.252	0.067	0.043	0.037	0.162
AM	MSH2	0.060	0.399	0.395	0.361	0.340	0.120	0.139
REVEL	BRCA1	0.161	0.429	0.336	0.285	0.268	0.058	0.140
REVEL	BRCA2	0.130	0.059	0.149	0.176	0.071	0.025	0.111
REVEL	BAP1	0.213	0.000	0.000	0.092	0.213	0.149	0.138
REVEL	RAD51C	0.259	0.000	0.000	0.143	0.259	0.185	0.187
REVEL	MSH2	0.060	0.701	0.577	0.485	0.642	0.106	0.329
ESM-2	BRCA1	0.161	0.276	0.304	0.212	0.114	0.050	0.044
ESM-2	BRCA2	0.130	0.371	0.371	0.288	0.241	0.075	0.035
ESM-2	BAP1	0.213	0.312	0.327	0.171	0.121	0.086	0.092
ESM-2	RAD51C	0.259	0.579	0.538	0.246	0.320	0.122	0.102
ESM-2	MSH2	0.060	0.762	0.670	0.531	0.702	0.165	0.234

S5 Implementation

Inputs: ClinVar `variant_summary.txt.gz` of 26 October 2026 (SHA-256 in `data/MANIFEST.txt`); MaveDB score sets `urn:mavedb:00000097-0-2` (BRCA1), `00001279-a-1` (BRCA2), `00000662-0-1` (BAP1), `00000673-0-1` (RAD51C) and `00000050-a-1` (MSH2, positive scores = loss of function); Ensembl VEP REST on GRCh38 (release 116) with the AlphaMissense and REVEL annotations; ESM-2 `esm2_t33_650M_UR50D`. Distance from the class boundary is the absolute difference between a variant's assay score and the midpoint of the two class medians in its gene, divided by the distance between the medians. Expected calibration error

Table S3: Sensitivity to the assay-label rule. Each gene is relabelled with the authors' classes or with the control rule at the stated posterior cut-off; other genes are unchanged. Per-class: lowest mean per-class coverage over the three predictors (50 random halvings, $\alpha = 0.1$); abstention: range over predictors; bounds held: oracle selections whose certified bound the true likelihood ratio respected (100 draws per predictor); LoF-supp.: share of draws certifying LoF-supporting calls, range over predictors.

Gene	Labels	n	LoF	Per-class	Abstention	Bounds held	LoF-supp.
BRCA2	controls, 0.80	3,971	0.136	0.894	0.49–0.53	294/295	0.63–0.96
BRCA2	controls, 0.95 (main)	3,435	0.130	0.896	0.36–0.42	423/423	0.90–0.99
BRCA2	controls, 0.99	2,341	0.151	0.897	0.30–0.37	591/593	0.94–1.00
BAP1	authors (main)	4,294	0.213	0.901	0.39–0.47	460/460	0.90–1.00
BAP1	controls, 0.95	4,221	0.100	0.899	0.13–0.19	777/777	1.00
RAD51C	authors (main)	2,225	0.259	0.895	0.18–0.28	558/561	0.84–0.99
RAD51C	controls, 0.95	832	0.538	0.902	0.07–0.19	868/868	0.98–1.00

Table S4: Calls judged against ClinVar labels (exploratory). Thresholds ($\alpha = 0.1$) are set on each gene's assay labels; one ClinVar variant with $\geq$2 stars is kept per residue. n: kept P/LP and B/LB variants; calls: LoF (or functional) calls among P/LP / among B/LB variants; L^+, U^-: likelihood-ratio bounds from one-sided Clopper–Pearson endpoints at level 0.025 each (simultaneous level 0.95 per ratio). Pooled rows sum the kept counts over genes, each gene with its own thresholds. BAP1 and RAD51C have too few labelled residues to be informative.

Pred.	Gene	n P/LP	n B/LB	LoF calls	L^+	func. calls	U^-
AM	BRCA1	75	170	56/1	19.6	8/153	0.24
AM	BRCA2	60	210	49/8	9.44	6/164	0.29
AM	BAP1	6	8	5/0	0.97	1/8	1.02
AM	RAD51C	8	3	5/1	0.27	2/2	6.90
AM	MSH2	72	55	50/6	2.58	9/46	0.31
AM	pooled	221	446	165/16	11.9	26/373	0.21
REVEL	BRCA1	75	170	48/10	4.94	8/134	0.28
REVEL	BRCA2	60	210	44/9	7.56	11/182	0.37
REVEL	BAP1	6	8	4/0	0.60	2/8	1.23
REVEL	RAD51C	8	3	6/0	0.49	2/3	2.23
REVEL	MSH2	72	55	43/3	3.14	5/38	0.28
REVEL	pooled	221	446	145/22	7.99	28/365	0.23
ESM-2	BRCA1	75	170	47/1	15.7	10/159	0.26
ESM-2	BRCA2	60	210	33/4	8.66	9/184	0.32
ESM-2	BAP1	6	8	5/0	0.97	1/5	2.62
ESM-2	RAD51C	8	3	5/0	0.35	1/1	62.7
ESM-2	MSH2	72	55	44/3	3.23	3/33	0.25
ESM-2	pooled	221	446	134/8	15.4	24/382	0.19

uses ten equal-mass bins. AUROC intervals are percentiles of 1,000 bootstrap resamples of residues. The region split divides each gene's assayed residues into four contiguous blocks with equal numbers of distinct positions. ClinVar records whose aggregate germline classification is empty (somatic-only submissions) are treated as unclassified. All random draws use NumPy generators with fixed seeds (2027 plus a fixed offset per experiment). The experiments run on a laptop CPU with Python 3.13, NumPy 2.5, SciPy 1.18, scikit-learn 1.9 and pandas 3.1; ESM-2 scoring used fair-esm 2.0 and PyTorch 2.14 on an Apple-silicon GPU. Exact versions and data checksums are in the repository. The Platt map is checked to be increasing (positive slope) every time it is fitted, and every fit in the experiments passed. Certified bounds use the conventions of Section 3: $U^- = \infty$ when $\ell_{\mathrm{TN}} = 0$, an empirical ratio $x/0$ is ∞ for $x > 0$, and 0/0 is undefined (never a claim); boundary cases are covered by unit tests (`make test`). The VUS-shift LoF shares are residue-weighted for both the calibration sample (one variant per residue) and the VUS (each residue weighted equally, then its

variants).